

3 hours on examples

Problem 1

- a) $z^4 = -1; z^8 = 1$ therefore, the primitive 8th roots of unity.
 b) $\cos(5\theta) = \operatorname{Re}(\cos(5\theta) + i \sin(5\theta)) = \operatorname{Re}((\cos(\theta) + i \sin(\theta))^5)$

$$= \cos^5(\theta) - 10 \sin^2(\theta) \cos^3(\theta) + 5 \sin^4(\theta) \cos(\theta)$$

From here bash out using $\sin^2 \theta = 1 - \cos^2 \theta$.

- c) We may express $\cos(30\theta) = f(\cos(\theta))$ for some integer polynomial f (expand with binom as before). Therefore if $\cos(1)$ is rational, then $\cos(30)$ is also rational, which it is not.

Problem 4 (Z01B65DA)

We let the roots of unity be $a_1 + a_2 + \dots + a_7 = 0$. Note that it is necessary and sufficient for no subsets of size 1, 2 or 3 to sum to zero (since the complementary subset will then also sum to 0).

Roots of unity cannot be zero, so there will never be a subset with one element that sums to zero.

For us to have no subsets with two elements to sum to zero, we must obviously not have any element that is the negative of another.

For us to have no subsets of three elements to sum to zero, it is sufficient to not have a triplet of roots of unity that are at 120° angles with each other as all triplets that sum to zero satisfies this.

The only remaining requirement to deal with is to have the seven roots of unity sum to zero. We let $\omega := \exp\left(\frac{i\pi}{15}\right)$. Since $\omega^5 + \omega^{-5} = 0$,

$$0 = \omega^0 + \omega^6 + \omega^{12} + \omega^{18} + \omega^{24}$$

$$\Leftrightarrow 0 = \omega^0 + (\omega^1 + \omega^{11}) + \omega^{12} + (\omega^{13} + \omega^{23}) + \omega^{24}$$

5 clubs; 4.5 hours

Problem 5 (SINSUM)

We let $\omega = \exp\left(\frac{i\pi}{180}\right)$.

We have,

$$\begin{aligned} & \sin(1^\circ) + \sin(2^\circ) + \dots + \sin(179^\circ) \\ &= \frac{(e^{i(1^\circ)} + e^{i(2^\circ)} + \dots + e^{i(179^\circ)}) - (e^{i(-1^\circ)} + e^{i(-2^\circ)} + \dots + e^{i(-179^\circ)})}{2i} \\ &= \frac{\frac{e^{i(180^\circ)} - e^{i(1^\circ)}}{e^{i(1^\circ)} - 1} - \frac{e^{i(-180^\circ)} - e^{i(-1^\circ)}}{e^{i(-1^\circ)} - 1}}{2i} = \frac{\frac{-1 - e^{i(1^\circ)}}{e^{i(1^\circ)} - 1} - \frac{-1 - e^{i(-1^\circ)}}{e^{i(-1^\circ)} - 1}}{2i} \end{aligned}$$

$$= \frac{-2 - 2e^{i(1^\circ)}}{e^{i(1^\circ)} - 1} = \frac{-1 - e^{i(1^\circ)}}{ie^{i(1^\circ)} - i} = i \frac{1 + e^{i(1^\circ)}}{e^{i(1^\circ)} - 1}$$

Note that,

$$\tan(x^\circ) = \frac{\frac{e^{i(x^\circ)} - e^{i(-x^\circ)}}{2i}}{\frac{e^{i(x^\circ)} + e^{i(-x^\circ)}}{2}} = i \frac{e^{i(x^\circ)} - e^{i(-x^\circ)}}{-e^{i(x^\circ)} - e^{i(-x^\circ)}} = i \frac{e^{i(2x^\circ)} - 1}{-e^{i(2x^\circ)} - 1}$$

We therefore need,

$$\begin{aligned} i \frac{e^{i(2x^\circ)} - 1}{-e^{i(2x^\circ)} - 1} &= i \frac{1 + e^{i(1^\circ)}}{e^{i(1^\circ)} - 1} \\ \Leftrightarrow \frac{e^{i(2x^\circ)} - 1}{-e^{i(2x^\circ)} - 1} &= \frac{1 + e^{i(1^\circ)}}{e^{i(1^\circ)} - 1} \\ \Leftrightarrow e^{i(2x+1^\circ)} - e^{i(2x^\circ)} - e^{i(1^\circ)} + 1 &= -e^{i(2x+1^\circ)} - e^{i(2x^\circ)} - e^{i(1^\circ)} - 1 \\ \Leftrightarrow e^{i(2x+1^\circ)} &= -1 \end{aligned}$$

This is accomplished for $x = \frac{179}{2}^\circ$.

Problem 6 (23AIMEII8)

$$= (\omega^0 + \omega^0 + \omega^0)(\omega^3 + \omega^1 + \omega^0)(\omega^6 + \omega^2 + \omega^0)(\omega^2 + \omega^3 + \omega^0)(\omega^5 + \omega^4 + \omega^0)(\omega^1 + \omega^5 + \omega^0)(\omega^4 + \omega^6 + \omega^0)$$

Noting that $(\omega^6 + \omega^2 + \omega^0)(\omega^5 + \omega^4 + \omega^0) = \frac{(\omega^3 + \omega^1 + \omega^0)}{\omega} \cdot \frac{(\omega^3 + \omega^1 + \omega^0)}{\omega^3}$ and $(\omega^2 + \omega^3 + \omega^0)(\omega^1 + \omega^5 + \omega^0)$ is the conjugate of that value, the product is,

$$\begin{aligned} &= 3(\omega^3 + \omega^1 + \omega^0)^3(\omega^4 + \omega^6 + \omega^0)^3 \\ &= 3(3 + \omega^2 + \omega^3 + \omega^5 + \omega + \omega^6 + \omega^4)^3 \\ &= 24 \end{aligned}$$

9 clubs; 5.5 hours

Problem 7 (12MP4G13)

The product is equal to zero if and only if one of the factors is equal to zero.

We claim that this occurs if and only if $n \equiv 3 \pmod{6}$. We first prove sufficiency. We let,

$$n := 6m + 3$$

We plug in $k = 2m + 1$ to get, $\left(1 + e^{\frac{2\pi i k}{n}}\right)^n + 1 = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^n + 1$. For $n \equiv 3 \pmod{6}$, this clearly evaluates to zero.

We now prove necessity.

$$\begin{aligned}\left(1 + e^{\frac{2\pi ik}{n}}\right)^n + 1 = 0 &\Leftrightarrow \left(1 + e^{\frac{2\pi ik}{n}}\right)^n = -1 \\ &\Rightarrow \left|1 + e^{\frac{2\pi ik}{n}}\right| = 1\end{aligned}$$

Since $\left|e^{\frac{2\pi ik}{n}}\right| = 1$, we have, $1 + e^{\frac{2\pi ik}{n}} \in \left\{\frac{1}{2} + \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i\right\}$. Looking at the argument, this means $n \equiv 3 \pmod{6}$ is necessary. Therefore, we have $n = 3, 9, \dots, 2007$. There are 335 solutions.

Problem 10 (15MP4G15)

We may rewrite this as,

$$\begin{aligned}(z^4 + 4z^3i - 6z^2 - 4zi + 1) + 8z + 8i &= 0 \\ \Leftrightarrow (z + i)^4 &= -8(z + i)\end{aligned}$$

We have one solution $z = -i$.

$$\Leftrightarrow (z + i)^3 = -8$$

This gives three more solutions that are equally spaced from each other and two units from $-i$. The answer is hence,

$$2 + 2 + 2 + 2\sqrt{3} + 2\sqrt{3} + 2\sqrt{3} = 6 + 6\sqrt{3}$$

15 clubs; 6.5 hours

Problem 12 (16MP4G18)

Consider the generating function,

$$F(x) = (x + 1)(x^2 + 1) \dots (x^{15} + 1)$$

We want the sum of coefficients of the terms with degree a multiple of 5. Using the roots of unity filter, where $\omega = \exp\left(\frac{2\pi i}{5}\right)$,

$$\begin{aligned}&\frac{F(1) + F(\omega) + F(\omega^2) + F(\omega^3) + F(\omega^4)}{5} \\ &= \frac{2^{15} + 4((\omega^0 + 1)(\omega^1 + 1)(\omega^2 + 1)(\omega^3 + 1)(\omega^4 + 1))^3}{5} \\ &= \frac{2^{15} + 4 \cdot 8}{5} = 6560\end{aligned}$$

Problem 13 (16MP4G20)

For any j , by multiplication,

$$\sum_{k=1}^5 \exp\left(\frac{a_k \pi i}{12}\right) = 0 \Leftrightarrow \sum_{k=1}^5 \exp\left(\frac{(a_k + j) \pi i}{12}\right) = 0$$

It's symmetric what the first choice is, so WLOG, let $a_1 = 0$. We count how many choices of the next four choices satisfy the conditions. We let unit vectors,

$$v_j := \exp\left(\frac{a_j \pi i}{12}\right)$$

Note that v_j are 24th roots of unities.

Case 1: Some non-empty proper subset of $v_1, \dots, v_5$ sum to zero.

Since a set of one unit vector cannot sum to zero alone, we must form a pair and a triplet each summing to zeroes.

Subcase 1a: $v_0 = 1$ is part of the pair.

There exists some $a_j = -1$.

If the triplet contains -1 , then a_2, a_3, a_4, a_5 are $(12, 12, 4, 20)$ in some order for 12 permutations.

Otherwise, a_2, a_3, a_4, a_5 are $(k, k + 8, k + 16, 12)$ for $0 \leq k < 8; k \neq 4$ in some order for $7 \cdot 24 = 168$ cases.

180 cases in subcase 1a.

Subcase 1b: $v_0 = 1$ is not part of a pair.

This means it is part of a triplet and nothing else. We have $8, 16 \in \{a_2, a_3, a_4, a_5\}$. If the pair contains $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, then there is a repeat so 12 permutations. Analogous 12 permutations for if it contains $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$. The other 9 cases possible each give 24 permutations.

240 cases in subcase 1b.

Case 2: No proper non-empty subsets of $v_1, \dots, v_5$ sum to zero.

We claim that there are no solutions.

Note that the 24th cyclotomic polynomial is,

$$\begin{aligned} & \left(x - \exp\left(\frac{2\pi i}{24}\right)\right) \left(x - \exp\left(\frac{10\pi i}{24}\right)\right) \left(x - \exp\left(\frac{14\pi i}{24}\right)\right) \left(x - \exp\left(\frac{22\pi i}{24}\right)\right) \left(x - \exp\left(\frac{26\pi i}{24}\right)\right) \left(x - \exp\left(\frac{34\pi i}{24}\right)\right) \left(x - \exp\left(\frac{38\pi i}{24}\right)\right) \left(x - \exp\left(\frac{46\pi i}{24}\right)\right) \\ & = X^8 - X^4 + 1 \end{aligned}$$

Note that $F(X) = X^{a_1} + X^{a_2} + X^{a_3} + X^{a_4} + X^{a_5}$ is a multiple of this, plugging in X for primitive roots of unities. Since there are no pairs of terms in X that sum to zero, we may take $F(X) \bmod X^{12}$ to have,

$$F(X) = c_0 + c_1X + \dots + c_{11}X^{11}$$

For $\sum |c_i| = 5$. Writing,

$$F(X) = (X^8 - X^4 + 1)G(X)$$

Since $G(X)$ has degree less than 4, $c_0 = -c_4 = c_8$; $c_1 = -c_5 = c_9$; This means that $3 \mid \sum |c_i|$. (!)

In total, we have $\frac{420}{24^3} = \frac{35}{12 \cdot 24 \cdot 24} = \frac{35}{27648}$.

Problem 14 (17MP4G19)

We consider the sides of the 13-gon to be distinct unit vectors $v_0, v_1, \dots, v_{12}$ on the complex circle that sum to zero. WLOG, let $v_0 = 1$. Since the internal angles are all multiples of 20° , $v_0, \dots, v_{12}$ must be distinct 18th roots of unities. We denote $\omega := \exp\left(\frac{i\pi}{9}\right)$.

Since stuff like $\cos(20^\circ)$, $\cos(40^\circ)$, $\cos(80^\circ)$ are irrational, the only way we can incorporate them and end with 0 is to add them in pairs that add to zero or triplets spaced 120° apart from each other that adds to zero.

Finally, to cancel out $v_0 = 1$, we must use $\omega^6 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\omega^{12} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$. The construction is as follows:

$$v_0 = 1; v_1 = \omega; v_2 = \omega^2; v_3 = \omega^4; v_4 = \omega^5; v_5 = \omega^6; v_6 = \omega^8; v_7 = \omega^{10}; v_8 = \omega^{11}; v_9 = \omega^{12}; v_{10} = \omega^{13}; v_{11} = \omega^{14}; v_{12} = \omega^{17}$$

The angles are hence,

$$160^\circ, 160^\circ, 140^\circ, 160^\circ, 160^\circ, 140^\circ, 140^\circ, 160^\circ, 160^\circ, 160^\circ, 160^\circ, 120^\circ, 160^\circ$$

32 clubs; 8 hours

Problem 16 (18MP4G18)

Since $31 \equiv 3 \pmod{4}$, by FCT $k^2 \equiv -1 \pmod{31}$ is impossible. Therefore, $\{0^2, 1^2, \dots, 15^2\} \pmod{31}$ is all the 16 quadratic residues modulo 31. This also means that,

$$\{-1^2, -2^2, \dots, -15^2\} \pmod{31}$$

Are all the 15 non-quadratic residues modulo 31. Hence,

$$\left| \prod_{k=0}^{15} \left(1 + e^{\frac{2\pi i k^2}{31}} \right) \right|^2 = \prod_{k=0}^{15} \left(1 + e^{\frac{2\pi i k^2}{31}} \right) \left(1 + e^{\frac{-2\pi i k^2}{31}} \right) = (1 + e^0) \cdot \prod_{k=0}^{31} \left(1 + e^{\frac{2\pi i k}{31}} \right)$$

The latter product is 2. Therefore,

$$\left| \prod_{k=0}^{15} \left(1 + e^{\frac{2\pi i k^2}{31}} \right) \right|^2 = 4 \Leftrightarrow \left| \prod_{k=0}^{15} \left(1 + e^{\frac{2\pi i k^2}{31}} \right) \right| = 2$$

Problem 17 (19MP4G16)

We employ complex coordinates with $A = \frac{1}{2}$; $G = -\frac{1}{2}$. We let,

$$\overrightarrow{GA} = 1; \overrightarrow{AB} = \exp\left(\frac{2\pi i}{7}\right); \overrightarrow{BC} = \exp\left(\frac{4\pi i}{7}\right); \dots$$

$$B = \frac{1}{2} + e^{\frac{2i\pi}{7}};$$

$$C = \frac{1}{2} + e^{\frac{2i\pi}{7}} + e^{\frac{4i\pi}{7}};$$

$$P = C + \overrightarrow{DE} = \frac{1}{2} + e^{\frac{2i\pi}{7}} + e^{\frac{4i\pi}{7}} + e^{\frac{8i\pi}{7}}$$

Since the midpoint of AG is the origin, the desired quantity is simply $|P|$. Note that,

$$\begin{aligned} |P|^2 &= \left(\frac{1}{2} + e^{\frac{2i\pi}{7}} + e^{\frac{4i\pi}{7}} + e^{\frac{8i\pi}{7}}\right) \left(\frac{1}{2} + e^{\frac{12i\pi}{7}} + e^{\frac{10i\pi}{7}} + e^{\frac{6i\pi}{7}}\right) \\ &= \frac{1}{2} \left(\frac{1}{2} + e^{\frac{2i\pi}{7}} + e^{\frac{4i\pi}{7}} + e^{\frac{8i\pi}{7}} + e^{\frac{12i\pi}{7}} + e^{\frac{10i\pi}{7}} + e^{\frac{6i\pi}{7}}\right) + e^{\frac{14i\pi}{7}} + e^{\frac{12i\pi}{7}} + e^{\frac{8i\pi}{7}} + e^{\frac{16i\pi}{7}} + e^{\frac{14i\pi}{7}} + e^{\frac{10i\pi}{7}} \\ &\quad + e^{\frac{20i\pi}{7}} + e^{\frac{18i\pi}{7}} + e^{\frac{14i\pi}{7}} \\ &= -\frac{1}{4} + 2 = \frac{7}{4} \end{aligned}$$

Therefore, the answer is $\frac{\sqrt{7}}{2}$.

42 clubs; 9 hours

Problem 19 (08PTNMA5)

FTSOC, let $\max(\deg(f), \deg(g)) < n - 1$. We have (m, n) be the coordinates of the center of the regular n -gon. We let,

$$h(x) := f(x) + ig(x) - (m + ni)$$

Note that h is a complex polynomial with $\deg(h) < n - 1$. We may plug in $x = 0, \dots, n - 1$ to see that $h(0), h(1), \dots, h(n - 1)$ are the direction vectors from the center of the n -gon to the vertices. We let,

$$p(x) := \frac{h(x)}{h(0)}$$

Plugging in $x = 1, \dots, n - 1$, we see that $p(0) = 1; p(1) = \exp\left(\frac{2\pi i}{n}\right); \dots; p(n - 1) = \exp\left(\frac{2\pi i(n-1)}{n}\right)$. We finally let,

$$q(x) := p(x+1) - \exp\left(\frac{2\pi i}{n}\right)p(x)$$

Note that q has $n-1$ roots at $x = 1, \dots, n-1$. But this is impossible as $\deg(q) < n-1$. (!)

Problem 20 (90IMO6)

We consider each side as unit vectors on the complex plane. We want, for some permutation $(a_1, \dots, a_{1990})$ of $(1, \dots, 1990)$,

$$0 = \sum_{j=1}^{1990} a_j^2 \exp\left(\frac{2j\pi i}{1990}\right)$$

Note that as $1990 \equiv 2 \pmod{4}$, every 1990^{th} root of unity is in the same or directly opposite direction as a 995^{th} root of unity. We may hence write, for some possibly negative numbers $c_1, \dots, c_{995}$.

$$\sum_{j=1}^{1990} a_j^2 \exp\left(\frac{2j\pi i}{1990}\right) = \sum_{j=1}^{995} c_j \exp\left(\frac{2j\pi i}{995}\right)$$

We place $1^2, 2^2$ on opposite sides, $3^2, 4^2$ on opposite sides and so on and in such a way that all c_j are positive. Since $(2k)^2 - (2k-1)^2 = 4k-1$, we see that $(c_1, \dots, c_{995})$ is a permutation of $(3, 7, \dots, 3979)$. Notice that $\sum_{j=1}^{995} \exp\left(\frac{2j\pi i}{995}\right) = 0$. Therefore, we increment each c_j by one,

$$\sum_{j=1}^{995} c_j \exp\left(\frac{2j\pi i}{995}\right) = 0$$

Where $(b_1, \dots, b_{995})$ is a permutation of $(1, \dots, 995)$,

$$\Leftrightarrow \sum_{j=1}^{995} 4b_j \exp\left(\frac{2j\pi i}{995}\right) = 0$$

$$\Leftrightarrow \sum_{j=1}^{995} b_j \exp\left(\frac{2j\pi i}{995}\right) = 0$$

We let ω be a primitive 995^{th} root of unity. We let, $z := \omega^0 + 2\omega^{199} + 3\omega^{398} + 4\omega^{597} + 5\omega^{796}$. Adding $0 = \omega^0 + \omega^{199} + \omega^{398} + \omega^{597} + \omega^{796}$. Therefore,

$$z = 6\omega^0 + 7\omega^{199} + 8\omega^{398} + 9\omega^{597} + 10\omega^{796}$$

...

$$= 991\omega^0 + 992\omega^{199} + 993\omega^{398} + 994\omega^{597} + 995\omega^{796}$$

Note that, since $1 + \omega + \omega^2 + \dots + \omega^{198} = 0$,

$$z(1 + \omega + \omega^2 + \dots + \omega^{198}) = 0$$

$$\Leftrightarrow (\omega^0 + 2\omega^{199} + 3\omega^{398} + 4\omega^{597} + 5\omega^{796}) + (6\omega^1 + 7\omega^{200} + 8\omega^{399} + 9\omega^{598} + 10\omega^{797}) + \dots = 0$$

56 clubs; 11 hours